

NICHOLAS SOBCHAK

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[LinkedIn](#) | [GitHub](#) | U.S. Citizen, Eligible for U.S. Security Clearance

EXPERIENCE

Founding Software Developer – ZipTip

Dec 2025 – Present

Remote / Westminster, MD

- > Built and developed lambda functions for user authentication, account creation & management, payment processing, and data retrieval.
- > Created authentication and authorization services for ZipTip and Stripe accounts using Cognito.
- > Integrated RESTful APIs for communication between frontend and backend services using boto3.
- > Built unit & integration tests using pytest, automated AWS SAM deployment for prod and dev environments.

SKILLS

C++: Crow, Catch2, SQLite, JSONL

Python: AWS (Lambda, SAM CLI), boto3, Cognito, pytest, JSON

JavaScript: Node.js, Vite, ESLint, Prettier

Databases: SQL, DynamoDB

Tools: GitHub, Linux, Vim, Tmux, Docker, CMake, vcpkg, Valgrind, Clang (Tidy/Format)

TECHNICAL PROJECTS [GitHub](#)

QuickQuill – C++17, Crow, SQLite3, Catch2; JavaScript, Vite; Docker

Dec 2025 – Present

- > Built a 1.2 million word dictionary API with a Trie-based backend in C++, supporting a sub-XXms autocomplete feature. Created a RESTful API and implemented logging using Crow framework [QuickQuill](#).
- > Implemented in-memory caching, reducing repeated query latency from ~200ms to <0.18ms. Prevented race conditions with `std::mutex` and `std::scoped_lock`.
- > Made the Trie & Database backend memory-safe with `std::unique_ptr`, and handled Crow multithreading by managing database access with `thread_local` for thread safety.
- > Created unit & integration tests using Catch2 framework, automated in a CI/CD pipeline (GitHub Actions).
- > Dockerized and deployed the application on google cloud vps for persistent hosting reverse proxy with nginx.

LEADERSHIP AND INTERESTS

Ice Hockey > Team Captain; Two-time State Champion; Two-time National Qualifier

Orchestra > Section Leader; demonstrated collaboration, discipline, and adaptability

EDUCATION

University of Maryland, Baltimore County

Aug 2025 – May 2027

Bachelor of Science in Computer Science (Expected)

Carroll Community College

Sept 2023 – May 2025

Associate of Arts in Computer Science, GPA: 3.6 Overall, 4.0 Major